

Background Needed to Read *Post-Selection Inference via Algorithmic Stability*

Below is the background you need **before** reading *Post-Selection Inference via Algorithmic Stability*. I am focusing on material the paper assumes or uses quickly, not what it explains in detail.

0. The paper's one-sentence problem

Classical confidence intervals assume the target was fixed before seeing the data. This paper studies what happens when you first use the data to choose the target, then use the same data to infer that target. The paper says this coupling invalidates classical guarantees and proposes randomized, stable selection as a correction mechanism.

The mental model:

$$\text{data } y \longrightarrow \text{choose target } \hat{S}(y) \longrightarrow \text{construct CI for } \beta_{\hat{S}(y)}.$$

The issue is that $\hat{S}(y)$ depends on y , so the inference target is random and statistically entangled with the same noise used for estimation.

1. Basic probability prerequisites

You need to be comfortable with random variables, conditional probability, and probability inequalities.

Random variables and distributions

A random variable is not just a number. It is a function from outcomes to values. In this paper:

$$y \in \mathbb{R}^n$$

is the observed data vector, often modeled as

$$y \sim N(\mu, \sigma^2 I).$$

This means each coordinate of y is centered around an unknown mean vector μ , with Gaussian noise.

The paper often distinguishes between:

$$y$$

the actual observed random data, and

$$\omega$$

a fixed possible realization of the data. This matters because stability is defined by asking: “If the data had been ω , how different would the randomized algorithm’s output distribution be?”

Conditional probability

You need the idea that probabilities can change after conditioning on an event.

For example, before selection:

$$y_i \sim N(\mu_i, \sigma^2).$$

But after selecting

$$i^* = \arg \max_i y_i,$$

the selected value y_{i^*} is biased upward. You chose it because it was large. So

$$y_{i^*} - \mu_{i^*}$$

is not distributed like ordinary Gaussian noise anymore.

This is the basic winner’s curse.

Union bound

The union bound appears constantly:

$$P\left(\bigcup_{i=1}^d A_i\right) \leq \sum_{i=1}^d P(A_i).$$

This is the basis of Bonferroni corrections. If you want all d intervals to be valid simultaneously with total failure probability α , you can give each one failure probability α/d .

Tail bounds and quantiles

You should know:

$$z_{1-\alpha}$$

means the $(1 - \alpha)$ -quantile of a standard normal:

$$P(Z \leq z_{1-\alpha}) = 1 - \alpha.$$

So a two-sided Gaussian interval is usually:

$$y_i \pm z_{1-\alpha/2}\sigma.$$

When the paper writes things like

$$z_{1-\delta e^{-\eta/2}},$$

it is just making the nominal confidence level more conservative.

2. Classical confidence intervals

A confidence interval procedure $CI^{(\alpha)}$ has coverage $1 - \alpha$ if

$$P(\theta \in CI^{(\alpha)}) \geq 1 - \alpha.$$

Equivalently,

$$P(\theta \notin CI^{(\alpha)}) \leq \alpha.$$

Important: this probability is over repeated data draws. The parameter θ is fixed. The interval is random.

The paper assumes that for every fixed selection S , you already know how to build a valid classical interval:

$$P(\beta_S \notin CI_S^{(\alpha)}) \leq \alpha.$$

The question is: what if S is replaced by a data-dependent $\hat{S}(y)$?

That is the whole post-selection problem.

3. Why post-selection inference fails

The failure is not subtle. If you test or estimate many things and then report only the most interesting one, the reported result is biased.

Simple example

Suppose

$$y_1, \dots, y_d \sim N(0, 1)$$

independently, and all true means are zero.

If you select

$$i^* = \arg \max_i y_i,$$

then y_{i^*} is usually positive and often large.

A naive confidence interval

$$y_{i^*} \pm 1.96$$

will not have 95% coverage for $\mu_{i^*} = 0$, because y_{i^*} was selected for being extreme.

That is the winner's curse. The paper uses this as one of its motivating examples.

4. Selection target versus estimator

This distinction is essential.

In ordinary inference, you estimate a fixed parameter:

$$\theta.$$

In post-selection inference, the parameter itself depends on the selected model:

$$\beta_{\hat{S}}.$$

For linear regression, if M is a selected subset of variables, then the target is not necessarily the “true coefficient” from a true linear model. It is the population least-squares projection coefficient for that selected model:

$$\beta_M = X_M^+ \mu.$$

That means:

$$\beta_{\hat{M}} = X_{\hat{M}}^+ \mu$$

is random because $\hat{M}$ is random.

This is a major conceptual point. They are not necessarily doing inference for some fixed true sparse vector. They are doing inference for the projection target induced by the selected model.

5. Fixed-target inference versus random-target inference

For fixed S :

$$P(\beta_S \in CI_S) \geq 1 - \alpha.$$

But this does **not** imply:

$$P(\beta_{\hat{S}} \in CI_{\hat{S}}) \geq 1 - \alpha.$$

The invalid step would be treating $\hat{S}$ as if it were fixed. It is not. It is a function of the data.

A lot of the paper is about recovering something close to fixed-target inference by showing that the randomized selection $\hat{S}(y)$ behaves almost like an oracle selection $\hat{S}_0$ that does not depend on the realized data.

6. Data splitting

Data splitting is the standard simple fix.

Split data into two independent parts:

$$D = D_{\text{select}} \cup D_{\text{infer}}.$$

Use D_{select} to choose the target:

$$\hat{S} = \hat{S}(D_{\text{select}}).$$

Use D_{infer} to construct the interval.

Because D_{infer} is independent of $\hat{S}$, classical inference is valid conditional on the selected target.

The downside: you throw away information. Selection uses less data, inference uses less data, and independence may be impossible with temporal, spatial, or structured dependence. The paper emphasizes this comparison explicitly: data splitting is broad but can be less powerful and can require independence, while stability can work without constructing independent splits.

For your research context, this is probably the key motivation: **algorithmic stability may replace data splitting as a way to control adaptivity without wasting a full holdout set.**

7. Simultaneous inference and Bonferroni

Sometimes you want one interval to be valid. Sometimes you want many intervals to be valid simultaneously.

For selected model M , simultaneous coverage means:

$$P(\beta_{j \cdot M} \in CI_{j \cdot M} \text{ for all } j \in M) \geq 1 - \alpha.$$

Bonferroni handles this by giving each selected coordinate failure probability roughly:

$$\frac{\alpha}{|M|}.$$

That is why the paper's linear regression intervals include terms like:

$$z_{1-\delta/(2|\hat{M}|e^\eta)}.$$

The $|\hat{M}|$ handles multiple selected coefficients. The e^η handles selection adaptivity.

8. Linear regression background

You need the following linear algebra for the model-selection sections.

Fixed design regression

The paper treats $X \in \mathbb{R}^{n \times d}$ as fixed, not random, in the main regression setup.

The outcome is:

$$y \in \mathbb{R}^n.$$

For a selected model $M \subseteq [d]$, X_M is the submatrix containing only columns in M .

Least squares estimator

For fixed M , the ordinary least squares estimate is:

$$\hat{\beta}_M = (X_M^\top X_M)^{-1} X_M^\top y = X_M^+ y.$$

The population target is:

$$\beta_M = X_M^+ \mu.$$

Again, this does not require the linear model to be exactly correct. It is a projection target.

Variance of OLS coefficients

If

$$y \sim N(\mu, \sigma^2 I),$$

then for fixed M ,

$$\hat{\beta}_M - \beta_M = X_M^+ (y - \mu).$$

So

$$\hat{\beta}_M - \beta_M \sim N\left(0, \sigma^2 (X_M^\top X_M)^{-1}\right).$$

The standard error for coefficient j is:

$$\sigma \sqrt{((X_M^\top X_M)^{-1})_{jj}}.$$

This is why the paper's intervals look like:

$$\hat{\beta}_{j \cdot \hat{M}} \pm K \hat{\sigma}_{j \cdot \hat{M}}.$$

9. LASSO background

The paper uses LASSO as a selection method.

The constrained LASSO solves:

$$\min_{\theta} \frac{1}{2} \|y - X\theta\|_2^2 \quad \text{s.t.} \quad \|\theta\|_1 \leq C.$$

The penalized form is:

$$\min_{\theta} \frac{1}{2} \|y - X\theta\|_2^2 + \lambda \|\theta\|_1.$$

These are closely related. LASSO encourages sparse solutions, so the selected model is often:

$$\hat{M} = \{j : \hat{\theta}_j \neq 0\}.$$

The paper's stable LASSO is not ordinary LASSO. It is a randomized Frank-Wolfe-style approximation to constrained LASSO. The randomization is what gives stability.

You do not need to master LASSO theory to read the paper, but you need to know:

$$\ell_1 \text{ constraint} \Rightarrow \text{sparsity} \Rightarrow \text{model selection} \Rightarrow \text{post-selection inference problem}.$$

10. Marginal screening background

Marginal screening selects variables with the largest correlation with the outcome.

For feature column X_j , compute:

$$|X_j^\top y|.$$

Then select the top k variables.

This is simpler than LASSO. The paper uses it because it is easy to stabilize: add noise to the scores and select the largest noisy scores.

The stable version is roughly:

$$\hat{i} = \arg \max_i (|X_i^\top y| + \text{noise}_i).$$

The tradeoff is direct:

- more noise means more stability and narrower post-selection correction,
- but worse selection quality;
- less noise means better selection quality but larger correction.

This tradeoff is the paper's core design principle.

11. Differential privacy background

The paper borrows tools from differential privacy, but not for privacy. It uses privacy-style stability to control adaptivity.

Standard differential privacy says an algorithm's output distribution should not change much if one data point changes.

A simplified version:

$$P(A(D) \in O) \leq e^\epsilon P(A(D') \in O) + \delta$$

for neighboring datasets D, D' .

This paper uses a related but different notion:

$$P(Q \in O) \leq e^\eta P(W \in O) + \tau.$$

The paper calls this indistinguishability or max-divergence. It is the same flavor: two random outputs are close if every event has similar probability under both distributions.

You need to understand e^η :

- $\eta = 0$: almost identical distributions.
- small η : close distributions.
- large η : weaker control.

This paper's stability definition is explicitly rooted in differential privacy and adaptive data analysis.

12. Indistinguishability / max-divergence

This is the technical heart.

The paper writes:

$$Q \approx_{\eta, \tau} W$$

if for all measurable events O ,

$$P(Q \in O) \leq e^\eta P(W \in O) + \tau.$$

Interpretation:

Q cannot put much more probability on any event than W does, except for a multiplicative factor e^η and additive slack τ .

If $\tau = 0$, then:

$$P(Q \in O) \leq e^\eta P(W \in O).$$

This lets you transfer probability bounds from W to Q .

That is exactly what the main theorem does.

If a bad event has probability at most $\delta e^{-\eta}$ under the oracle world, then under the real selected world it has probability at most:

$$e^\eta(\delta e^{-\eta}) = \delta.$$

That cancellation is the entire reason the correction is $e^{-\eta}$.

13. The oracle selection idea

The oracle selection $\hat{S}_0$ is not an oracle that knows the true answer. It is a fictional random variable whose distribution approximates the randomized selection distribution but does not depend on the realized data y .

The paper's stability definition says:

$$A(y) \approx_{\eta, \tau} A_0$$

with high probability over y .

Interpretation:

After randomization, the selection output does not reveal too much about the particular dataset realization. So we can compare it to a selection that could have been generated without looking at the actual y .

This is the bridge:

$$(y, \hat{S}(y)) \approx (y, \hat{S}_0).$$

Since $\hat{S}_0$ is independent of y , classical inference works for $\hat{S}_0$. Stability transfers that validity back to $\hat{S}(y)$.

14. Randomization as stabilization

The paper repeatedly uses the same trick:

1. Compute a data-dependent score.
2. Add noise.
3. Select based on the noisy score.

Example:

$$\hat{i} = \arg \max_i (y_i + \xi_i),$$

where ξ_i is Laplace noise.

Why Laplace? Because Laplace distributions have density ratios that are easy to bound. If two scores differ by at most Δ , then adding Laplace noise with scale b gives likelihood ratio roughly:

$$\exp(\Delta/b).$$

So the stability parameter behaves like:

$$\eta \approx \frac{\text{sensitivity}}{\text{noise scale}}.$$

More noise means smaller η . Less noise means larger η .

15. Sensitivity

Sensitivity means: how much can the quantity you compute change when the data changes?

In this paper, the relevant changes are usually from replacing y by its mean μ , or by another typical realization.

For a linear score:

$$w^\top y,$$

if y is close to μ , then

$$|w^\top y - w^\top \mu|$$

is small with high probability.

That small deviation is the sensitivity-like quantity. The paper then calibrates noise to dominate this deviation.

General pattern:

$$\text{noise scale} \sim \frac{\text{typical score deviation}}{\eta}.$$

So if you want small η , you need large noise.

16. Laplace noise

The Laplace distribution with scale b has density:

$$f(x) = \frac{1}{2b} e^{-|x|/b}.$$

The key property:

$$\frac{f(x)}{f(x+c)} \leq e^{|c|/b}.$$

This is why Laplace noise is used in differential privacy and in this paper.

If changing the data shifts a score by at most c , then the noisy score distribution changes by at most a multiplicative factor $e^{c/b}$.

Set

$$b = c/\eta$$

and you get an e^η bound.

17. Composition

Composition means: if you run several stable procedures, the instability accumulates.

Simple version:

If one step is η -stable, then k steps are roughly $k\eta$ -stable.

This matters for:

- stable LASSO, which takes multiple optimization steps;
- marginal screening, which selects variables sequentially;
- running multiple selection algorithms before choosing a final model.

The paper says stability is robust under post-processing and composition, which is one of its practical advantages.

18. Post-processing

Post-processing means:

If $A(y)$ is stable, then any function of $A(y)$ is also stable.

For example, if the randomized LASSO coefficient vector is stable:

$$\hat{\theta}_{\text{LASSO}}$$

then its selected support is also stable:

$$\hat{M} = \{j : \hat{\theta}_j \neq 0\}.$$

This is useful because you can stabilize a rich object, then select models however you want from it.

19. Conditional versus marginal coverage

This paper mostly gives marginal, not conditional, guarantees.

Marginal coverage

$$P(\beta_{\hat{S}} \in CI_{\hat{S}}) \geq 1 - \alpha.$$

This averages over both data and selection.

Conditional coverage

$$P(\beta_S \in CI_S \mid \hat{S} = S) \geq 1 - \alpha \quad \text{for every } S.$$

Conditional coverage is stronger in one sense, but often causes very wide intervals because conditioning on the exact selection event can consume a lot of information.

The paper contrasts its approach with conditional selective inference, noting that conditional methods can overcondition and lead to wide intervals; randomized selective inference often requires expensive sampling, while the stability approach gives simple corrections.

For reading this paper, do not confuse:

$$P(\cdot)$$

with

$$P(\cdot \mid \hat{S} = S).$$

Most main results are unconditional/marginal.

20. PoSI background

PoSI means **P**ost-**S**election **I**nfERENCE.

The classical PoSI idea asks for intervals valid no matter which model selection procedure was used. This is very robust but conservative.

The paper's approach is less conservative because it does not protect against arbitrary selection. It protects against selection rules with a bounded stability parameter η .

So the hierarchy is:

$$\text{classical inference} < \text{stability-adjusted inference} < \text{fully selection-agnostic PoSI}$$

in terms of conservativeness.

Classical inference is narrow but invalid after selection. Fully general PoSI is valid but wide. Stability tries to interpolate between them.

21. Frank-Wolfe background

The stable LASSO algorithm uses a Frank-Wolfe-style method.

Frank-Wolfe solves constrained convex optimization problems:

$$\min_{\theta \in C} f(\theta).$$

Instead of projecting onto C , it repeatedly solves a linear minimization problem:

$$s_t = \arg \min_{s \in C} \langle s, \nabla f(\theta_t) \rangle$$

and updates:

$$\theta_{t+1} = (1 - \gamma_t)\theta_t + \gamma_t s_t.$$

For the ℓ_1 -ball, the extreme points are coordinate vectors:

$$\{\pm C e_j\}_{j=1}^d.$$

So each Frank-Wolfe step chooses a coordinate direction. The stable LASSO adds noise before choosing that direction.

You only need the high-level idea: **stable LASSO is iterative noisy coordinate selection inside an optimization algorithm.**

22. What the experiments measure

The paper's experiments compare stability to data splitting.

For LASSO, they measure things like false discovery rate and interval width.

For marginal screening, they measure average selection error:

$$\text{error} = \text{score of true top feature} - \text{score of selected feature}.$$

The experiments are not proving the theorem. The theorem gives validity. The experiments evaluate the utility tradeoff:

$$\text{more randomization} \Rightarrow \text{more stability} \Rightarrow \text{narrower correction} \Rightarrow \text{but worse selection}.$$

The empirical claim is that stable selection can often beat data splitting in selection quality for comparable interval width.

23. The key algebraic mechanism of the main theorem

You should understand this before reading Theorem 2.

Suppose the bad event is:

$$B = \{\beta_{\hat{S}} \notin CI_{\hat{S}}\}.$$

If real selection is close to oracle selection:

$$(y, \hat{S}(y)) \approx_{\eta, \tau + \nu} (y, \hat{S}_0),$$

then:

$$P_{\text{real}}(B) \leq e^\eta P_{\text{oracle}}(B) + \tau + \nu.$$

Since $\hat{S}_0$ is independent of y , classical inference gives:

$$P_{\text{oracle}}(B) \leq \delta e^{-\eta}.$$

Therefore:

$$P_{\text{real}}(B) \leq e^\eta \delta e^{-\eta} + \tau + \nu = \delta + \tau + \nu.$$

This is the whole theorem.

Everything else is about proving that specific selection algorithms are stable and quantifying the cost of making them stable.

24. How this connects to your broader research direction

The relevant abstraction is:

data reuse creates adaptivity bias.

Data splitting solves it by independence:

$$\hat{S}(D_1) \perp D_2.$$

Stability solves it by approximate independence:

$$(y, \hat{S}(y)) \approx (y, \hat{S}_0).$$

That distinction is probably the most important thing for your project.

If you are thinking about replacing inefficient data splitting in conformal or decision-risk pipelines, this paper gives one template:

1. Identify the adaptive object selected using data.
2. Randomize or regularize the selection rule.
3. Prove a stability parameter η .
4. Replace the split-sample correction with an η -dependent correction.
5. Quantify the utility loss from stabilization.

The hard part is not reading Theorem 2. The hard part is finding the right analogue of $\hat{S}$, $\hat{S}_0$, and η in your setting.